

SimpleMem: Efficient Lifelong Memory for LLM Agents

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SimpleMem: Efficient Lifelong Memory for LLM Agents

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- UNC Chapel Hill | UC Berkeley | UC Santa Cruz
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Background

Large Language Model (LLM) Agents

- Used in chatbots, assistants, autonomous agents
- Limited by **fixed context window**
- Cannot efficiently handle long multi-session conversations

Problem in Existing Systems

- Full-context storage → redundant information
- Iterative filtering → high token & computational cost
- Context inflation → degraded reasoning performance

Research Problem

Core Problem

How to design an **efficient lifelong memory system** for LLM agents that:

- Maintains long-term interaction history
- Removes redundant dialogue
- Preserves important semantic information
- Minimizes token usage during inference

Motivation

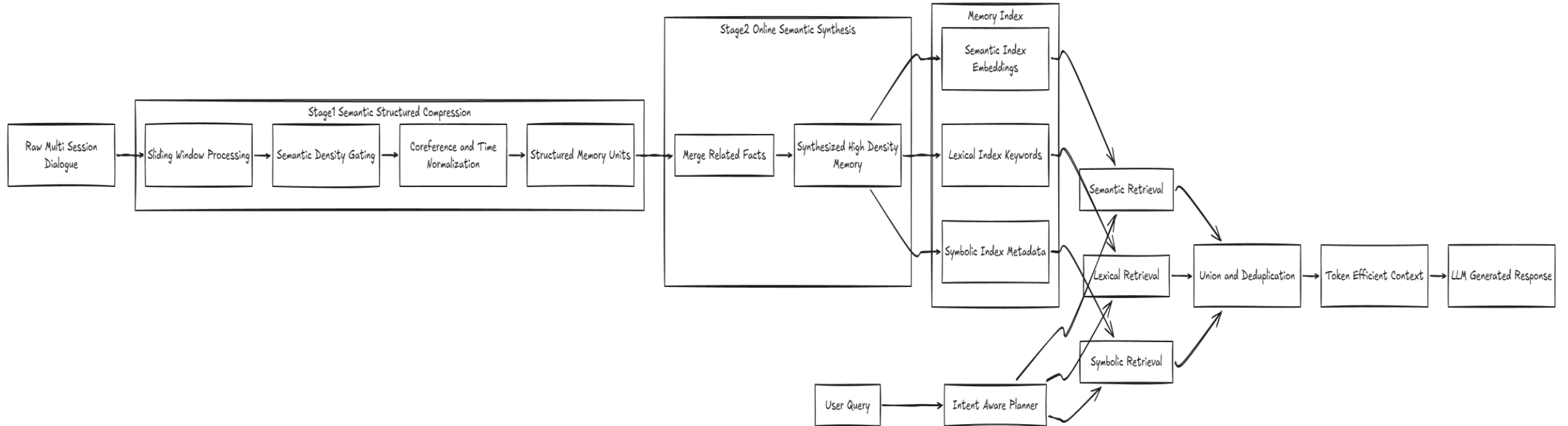
Future AI agents require:

- Multi-session reasoning
- Temporal understanding
- Personalized long-term memory
- Efficient computation

Existing approaches fail to balance:

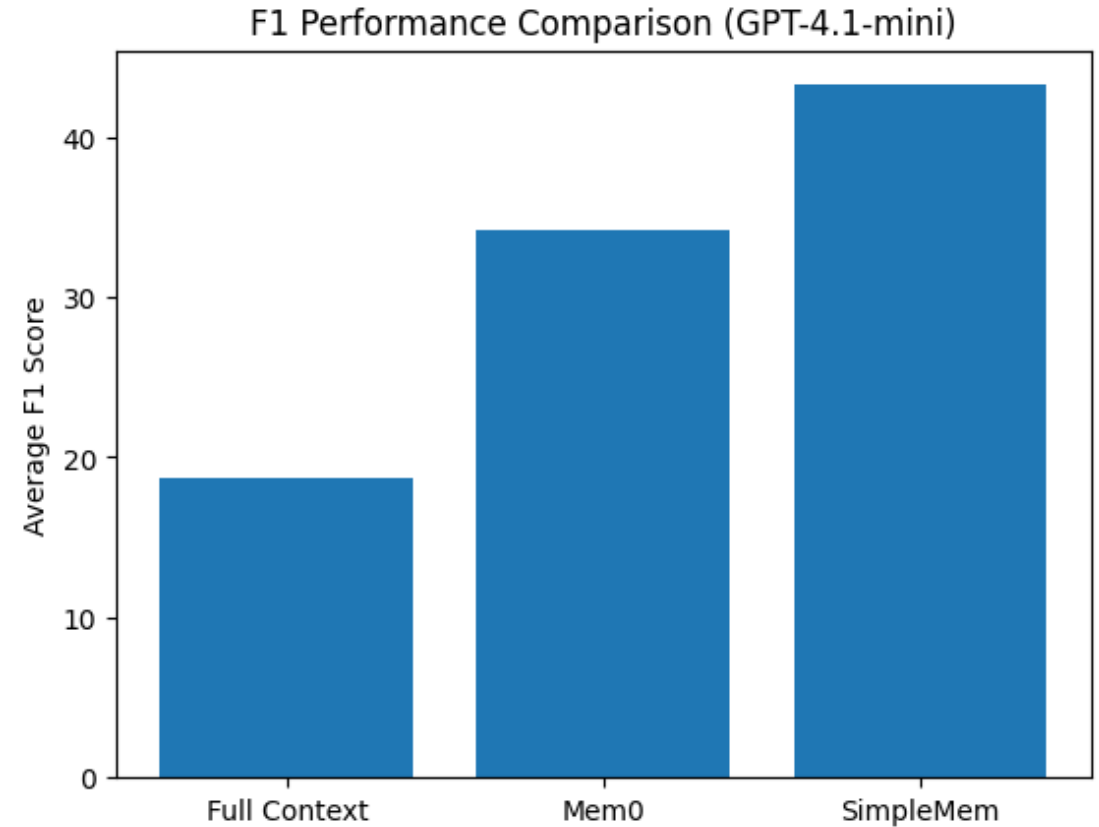
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SimpleMem Architecture



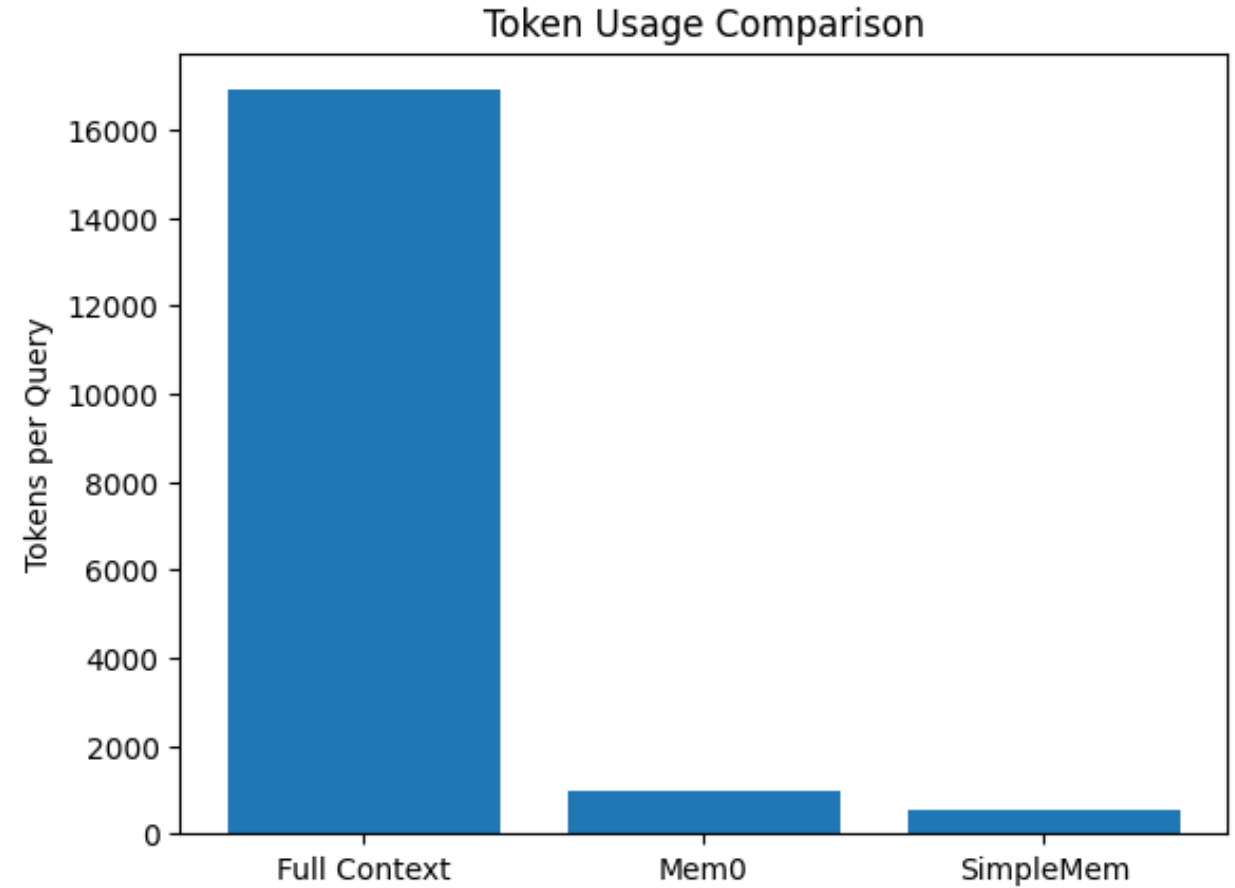
Performance Results (GPT-4.1-mini)

- Average F1 Score:
- Full Context: 18.70 | Mem0: 34.20 | SimpleMem: 43.24
- +26.4% improvement over Mem0



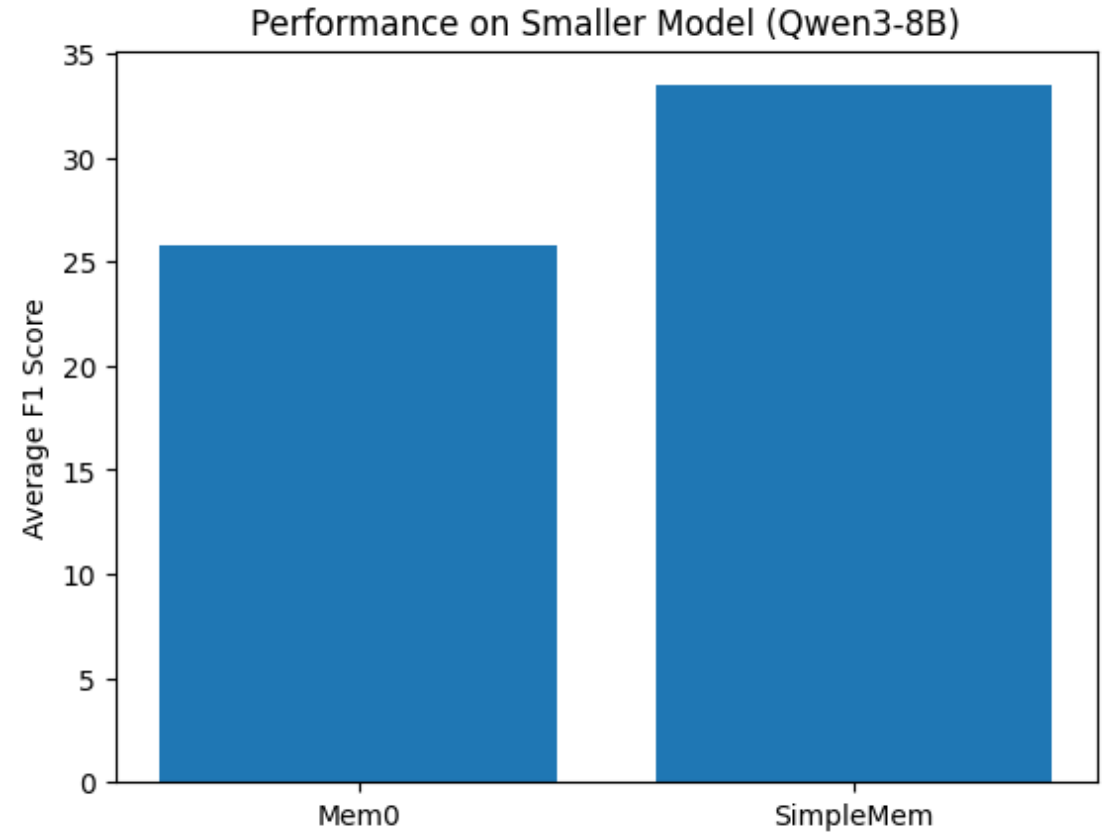
Token Efficiency

- Full-context systems: ~16,900 tokens/query
- SimpleMem: ~531 tokens/query
- ~30× reduction



Performance on Smaller Models

- Qwen3-8B: SimpleMem 33.45 F1 vs Mem0 25.80 F1
- Qwen2.5-1.5B: Strong performance despite low parameters
- SimpleMem enhances capability across all model sizes



Conclusion

- Filters low-density dialogue effectively
- Normalizes temporal references
- Synthesizes related information intelligently
- Uses adaptive multi-view retrieval
- Achieves state-of-the-art performance with 30× lower token cost